

AMRITA VISHWA VIDYAPEETHAM
SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DiffShield: A Tri-Modal Adversarial Framework for Robust Immunization Against Deepfake Generation

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Team Members		Guide	Project Type	Marks
CB.SC.U4CSE23437	Pranav Kishan T Y	Dr. Shanmuga Priya S. <i>Assistant Professor (Sl.Gd.), Dept. of CSE</i>	Type 1 Research	.../5
CB.SC.U4CSE23519	K Venkatesh			
CB.SC.U4CSE23536	Narain B K			
CB.SC.U4CSE23539	Karthikeyan P G			
CB.SC.U4CSE23556	Varun Raj V			

1 Project Proposal and Objectives

Project Idea: The proliferation of Latent Diffusion Models (LDMs) has democratized image synthesis but simultaneously enabled severe digital facial privacy violations via targeted deepfakes. Current adversarial immunization techniques are structurally fragile, operating exclusively in the single-modality pixel domain. We propose **Diff-Shield**, a Tri-Modal Adversarial Framework that mathematically immunizes facial images prior to public upload.

Project Objectives: The primary objectives to be achieved in this project are:

1. To formulate a Tri-Modal adversarial loss function that simultaneously attacks the Visual (VAE), Semantic (CLIP), and Structural (Canny) pathways of Stable Diffusion.
2. To ensure real-world robustness against social media compression by integrating an Expectation over Transformation (EoT) layer within the optimization loop.
3. To guarantee visual imperceptibility of the protective noise using L_∞ norm projection clamping.
4. To validate the framework's efficacy against state-of-the-art ControlNet geometric overrides, utilizing Explainable AI (Grad-CAM) to visualize the disruption.

2 Introduction and Background

Motivation: As generative AI capabilities accelerate, malicious actors increasingly utilize open-source frameworks like Stable Diffusion to generate non-consensual deepfakes. While initial defenses sought to disrupt the pixel latent space, attackers have adapted by using ControlNet to extract pristine facial geometry (via Canny edges) and textual inversion to force semantic likeness, completely bypassing pixel-level noise. There is a critical motivation to develop an architecture-agnostic defense that cripples the generative model's multi-modal understanding of the image.

Existing Literature & Gap Analysis: Existing solutions largely focus on isolated modalities (e.g., attacking only the VAE or only CLIP), which exhibit severe operational gaps when subjected to complex geometric overrides. A tabulated summary of key literature defining our research gap is presented below:

Paper & Reference	Proposed Method	Identified Limitation
PhotoGuard (Salman <i>et al.</i>) [1]	Attacks the VAE encoder latent space using PGD.	Easily washed off by standard JPEG compression and cropping.
AdvI2I (Naseer <i>et al.</i>) [3]	Targets CLIP textual embeddings for semantic disruption.	Fails against spatial conditioning tools like ControlNet.
Anti-DreamBooth [5]	Adds noise to disrupt model fine-tuning processes.	Computationally heavy; ineffective against zero-shot attacks.
DiffAM (Sun <i>et al.</i>) [6]	Adversarial makeup transfer to hide facial identity.	Leaves visible, unnatural artifacts on the user's face.
TUAP (Chen <i>et al.</i>) [4]	Targeted universal perturbations against CLIP vision models.	Single-modality; susceptible to test-time defenses [12].

3 Execution Plan

To successfully implement the DiffShield framework, the project execution is structured across four distinct software engineering and evaluation subdomains. This modular approach separates the data pipeline, the mathematical optimizer, the AI architectures, and the final validation metrics:

- **Subdomain 1: Data Processing & Transformation Pipeline**
 - *Component*: CelebA-HQ dataset ingestion and the Expectation over Transformation (EoT) layer.
 - *Learning/Execution Plan*: Implementing differentiable image processing pipelines via the `Kornia` library to apply JPEG compression and cropping augmentations during the forward pass, ensuring the adversarial noise is mathematically robust against real-world social media filters.
- **Subdomain 2: Adversarial Optimization Engine**
 - *Component*: The Projected Gradient Descent (PGD) mathematical loop and L_∞ projection clamping.
 - *Learning/Execution Plan*: Utilizing the PyTorch autograd engine to freeze model weights, calculate input-level gradients, and iteratively update the noise tensor. This explicitly includes writing custom min / max clamping functions to guarantee visual imperceptibility.
- **Subdomain 3: Multi-Modal Architecture Integration**
 - *Component*: The Tri-Modal Objective ($\mathcal{L}_{vis}, \mathcal{L}_{sem}, \mathcal{L}_{str}$).
 - *Learning/Execution Plan*: Integrating the HuggingFace `diffusers` library to synchronously load the pre-trained AutoencoderKL, the CLIP vision-language model, and a Canny edge detector. The focus is computing simultaneous MSE and Cosine Similarity errors across disparate latent spaces.
- **Subdomain 4: Robustness Benchmarking & Explainable AI (XAI)**
 - *Component*: Automated grading scripts and Grad-CAM visualizations.
 - *Learning/Execution Plan*: Building an inference pipeline to test poisoned images against SD 1.5 + ControlNet. This requires coding automated metric calculators (PSNR, SSIM, ASR) and extracting cross-attention gradients to plot Explainable AI heatmaps.

4 Fundamentals Learned in the Semester

The team built foundation via literature review of SOTA adversarial defenses. Key milestones include:

- **Single-Modality Defense Vulnerabilities**: Studying PhotoGuard [1] gave us precise understanding of PGD-based attacks on the VAE latent space, where the update rule $\delta_{t+1} = \Pi_{[-\epsilon, \epsilon]}(\delta_t + \eta \cdot \nabla_{\delta} \mathcal{L})$ enforces L_∞ imperceptibility. We diagnosed why it fails: since the perturbation is optimized solely in the VAE domain, standard JPEG compression is sufficient to wash it out entirely. Similarly, studying AdvI2I [3] and TUAP [4], which achieves 77.4% Attack Success Rate across 6 victim CLIP models by moving image embeddings towards adversarial targets text via cosine similarity minimization, revealed that single-modality semantic attacks collapse against spatial conditioning models like ControlNet. These gaps directly motivated our Tri-Modal approach.
- **Ensuring Real-World Robustness via EoT**: To prevent our perturbation from being eliminated by media compression, we studied Athalye *et al.* [10]. The core insight of Expectation over Transformation (EoT) is to replace standard single-input optimization with maximizing an expectation over a distribution of transformations $\mathcal{T}$: $\max_{\delta} \mathbb{E}_{t \sim \mathcal{T}}[\mathcal{L}(t(X + \delta))]$, sampling a new transformation at every forward-backward pass. We learnt that the transformations must be differentiable for gradients to flow through them, leading to our decision to implement the EoT layer using the `Kornia` library.
- **Taxonomy of Diffusion Attacks & Personalization Defenses**: We utilized the comprehensive survey by Zhang *et al.* [8] to map the landscape of text-to-image vulnerabilities, understanding how the VAE, UNet, and frozen CLIP text-encoder interact sequentially during inference and how ControlNet introduces a third, spatial conditioning pathway that latent defenses cannot address. Anti-DreamBooth [5] further clarified the mechanics of disrupting model fine-tuning, reinforcing the need for a zero-shot defense against geometric override attacks.
- **Explainable AI (XAI) for Phase 3 Validation**: To prepare for validation, we studied the application of Grad-CAM to adversarial settings via Chen *et al.* [13]. We learned that Grad-CAM computes spatial importance weights by pooling the gradients of a target score with respect to a chosen layer’s feature maps, producing a heatmap of the model’s focal regions. Applied to the cross-attention layers of the Stable Diffusion UNet, this functions as our primary XAI tool to visually document the scattering of the AI’s attention away from coherent facial features.

5 Architecture Diagram and Explanation of Design

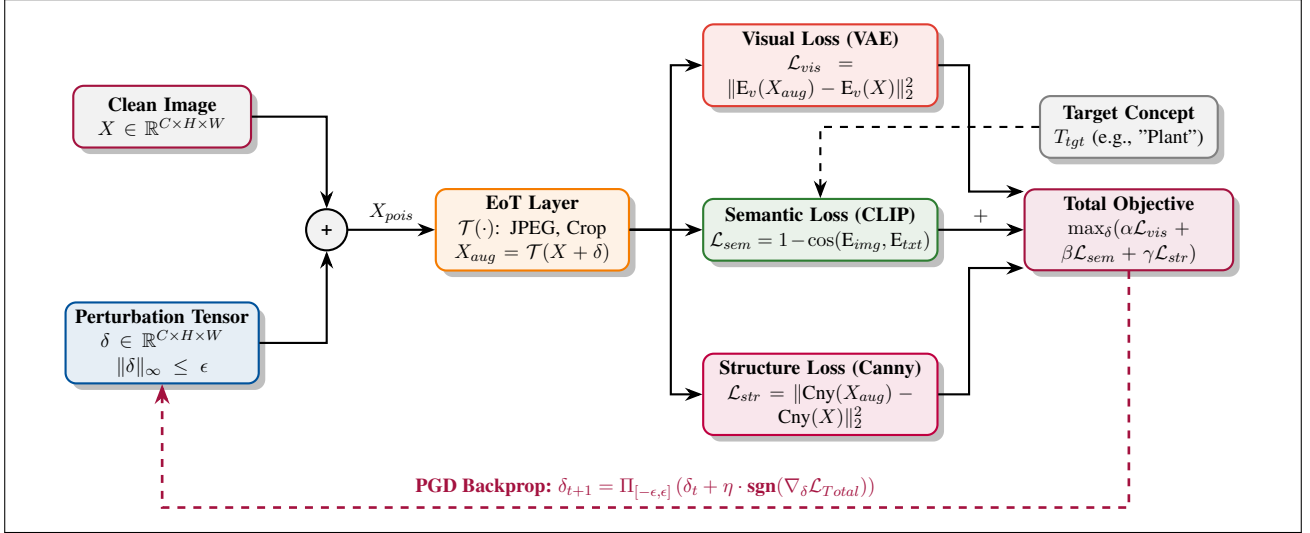


Figure 1: DiffShield Tri-Modal Iterative Optimization Pipeline

Module Breakdown and Design Justification:

- **Input & EoT Layer:** A noise tensor δ is injected into the clean image. Before optimization, the image passes through the Expectation over Transformation (EoT) module. *Design Choice:* We utilize differentiable augmentations (like JPEG compression) so the calculated gradients explicitly learn to survive the digital compression algorithms used by social media platforms.
- **Visual Loss Module ($\mathcal{L}_{vis}$):** Targets the VAE. *Design Choice:* By maximizing the Mean Squared Error (MSE) between the clean and poisoned images in the latent space, we destroy the spatial blueprint of the image before the diffusion UNet can process it.
- **Semantic Loss Module ($\mathcal{L}_{sem}$):** Targets the CLIP Vision Encoder. *Design Choice:* We maximize the Cosine Similarity toward a dynamically selected, orthogonal target concept (e.g., "a potted plant"). This explicitly forces the AI's language-vision bridge to misinterpret the human face as an inanimate object.
- **Structure Loss Module ($\mathcal{L}_{str}$):** Targets ControlNet geometry. *Design Choice:* By maximizing the high-frequency MSE of a Canny edge detector, the noise generates thousands of fake "micro-edges." This shatters the clean facial stencil into static, blinding ControlNet's geometric override capabilities.
- **Total Objective & PGD Loop:** The scaled total loss is backpropagated to update the noise tensor. *Design Choice:* We utilize the 'sign()' of the gradient to ensure even pixel perturbation, and apply an L_∞ projection (Π) of 8/255. This mathematically clamps the noise, guaranteeing human-eye imperceptibility.

6 Summary of Works Completed (Phase 1)

This semester, the team transitioned from problem identification to designing a mathematically validated architectural framework. Key milestones include:

- **Gap Formulation:** Analyzed SOTA defenses (e.g., PhotoGuard, AdvI2I), identifying their shared vulnerability to spatial overrides (ControlNet) and digital degradation to justify our Tri-Modal approach.
- **Mathematical Formulation:** Defined the joint-loss equations for the optimizer, including MSE for VAE Visual Collapse, Cosine Similarity for CLIP Semantic Erasure, and a differentiable Canny loss for Structural Sabotage.
- **Robustness Pipeline (EoT):** Designed an Expectation over Transformation layer using Kornia to ensure gradient calculation survives simulated real-world digital transformations (JPEG compression/resizing).
- **Constraint Mapping:** Finalized PGD as the core optimization engine and mapped the exact L_∞ norm boundary ($\epsilon = 8/255$) to mathematically guarantee visual stealth.

7 Plans for Upcoming Semesters

The subsequent phases of the project will be executed over the next two academic semesters, transitioning from theoretical formulation to software implementation and rigorous empirical evaluation.

Semester 7 (June 2026 to December 2026): Phase 2 Core Development

- **Environment Setup:** Configure the PyTorch and HuggingFace Diffusers environments on GPU instances and preprocess the CelebA-HQ dataset for the optimization loop.

- **Module Implementation:** Program the independent loss calculation scripts for the VAE autoencoder, the frozen CLIP text encoder, and the differentiable Canny edge detector.
- **PGD Pipeline Construction:** Develop the Projected Gradient Descent optimization loop, strictly enforcing the L_∞ mathematical projection to guarantee visual imperceptibility.
- **EoT Integration:** Embed the Expectation over Transformation layer using the Kornia library to apply differentiable JPEG compression and cropping during the forward pass.
- **Hyperparameter Tuning:** Conduct small-scale validation trials to balance the alpha, beta, and gamma weights, ensuring the gradient norms of all three modalities pull equally without causing vanishing gradients.

Semester 8 (December 2026 to April 2027): Phase 3 Evaluation and Documentation

- **Automated Benchmarking:** Execute large-scale simulated deepfake attacks by feeding the immunized images into Stable Diffusion 1.5 equipped with ControlNet spatial overrides.
- **Quantitative Analysis:** Calculate visual stealth metrics (PSNR, SSIM, LPIPS) and defensive robustness metrics (Attack Success Rate, CLIP-Score) across the test dataset.
- **Qualitative XAI Validation:** Generate Grad-CAM attention heatmaps to visually document the scattering of the AI cross-attention mechanisms.
- **Final Optimization:** Address edge-case failures and optimize the inference time of the perturbation generator.
- **Thesis Compilation:** Consolidate all experimental data, charts, and findings into the final project thesis and prepare a manuscript for potential research publication.

References

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